BELLABEAT: SMART DEVICE USAGE ANALYSIS

Google Data Analytics Capstone Case Study

AGENDA

- Project Overview & Goals
- Data & Analysis
- Recommendations
- Actionable Next Steps

PURPOSE

Analyze Fitbit smart device usage data to identify consumer trends and develop high-level marketing recommendations for the Bellabeat app

DATA SOURCE & SCOPE



The data analyzed was the Fitbit Fitness Tracker Data published on Kaggle by Mobius, originating from the Furberg et al. dataset covering March – May 2016 across two exports.



Combined and cleaned Daily Activity from both exports: 1,373 participant-day records and 35 distinct participant IDs.



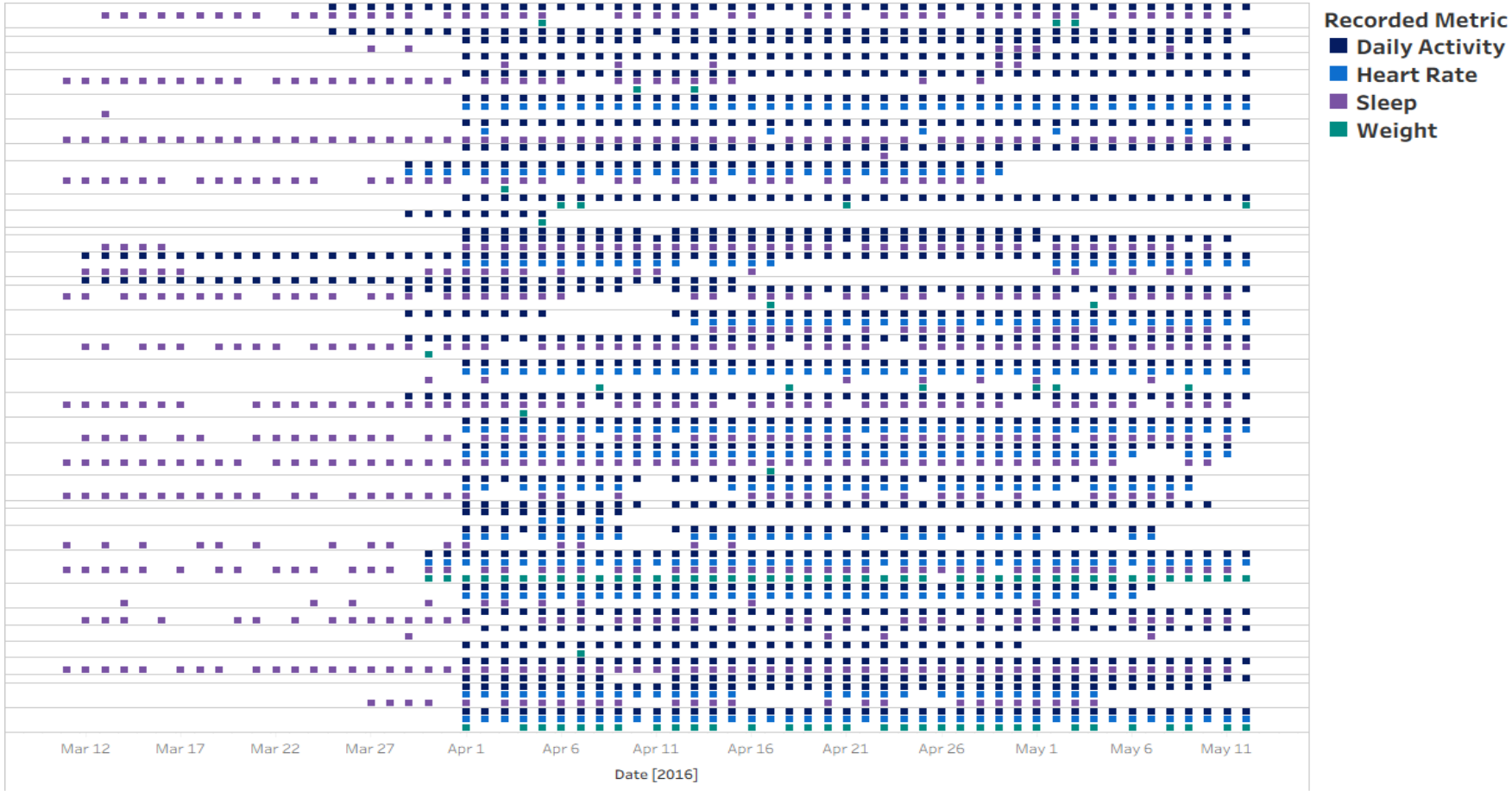
Combined and cleaned minute-level sleep data, reconstructed into 1,007 sleep sessions

DATASET LIMITATIONS

1. Data was collected in 2016 and may not reflect current behavior.
2. No demographic data, including age or gender, to establish relevance to Bellabeat customers.
3. Only 35 distinct IDs, with different participants contributing different amounts and types of records.
4. Cannot determine whether absent records reflect device capabilities, participant preferences, nonuse, or other factors.
5. Published description reports 30 eligible users, while our combined data contains 35 distinct IDs.

PARTICIPANT DATA COVERAGE

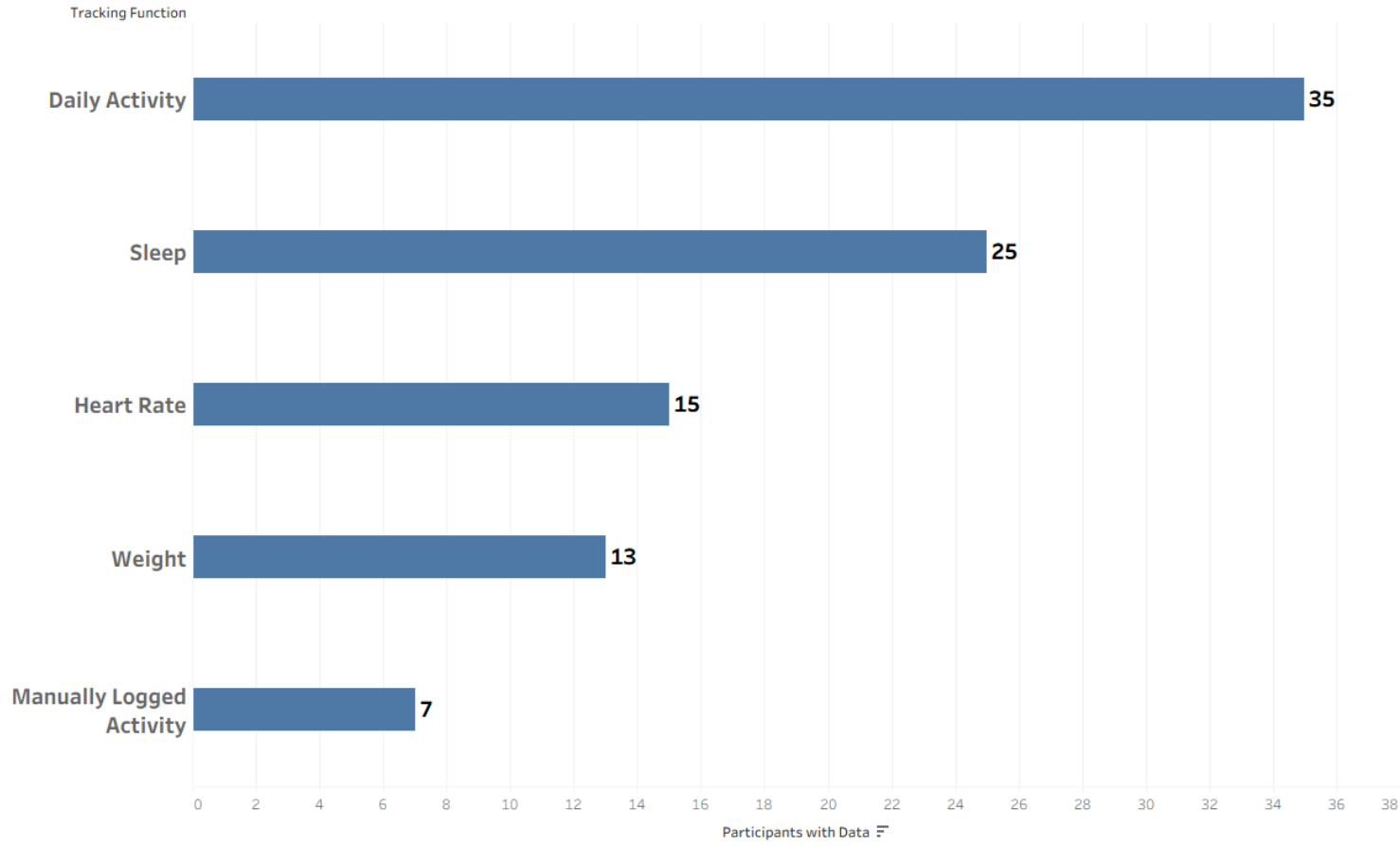
Horizontal groupings represent one participant



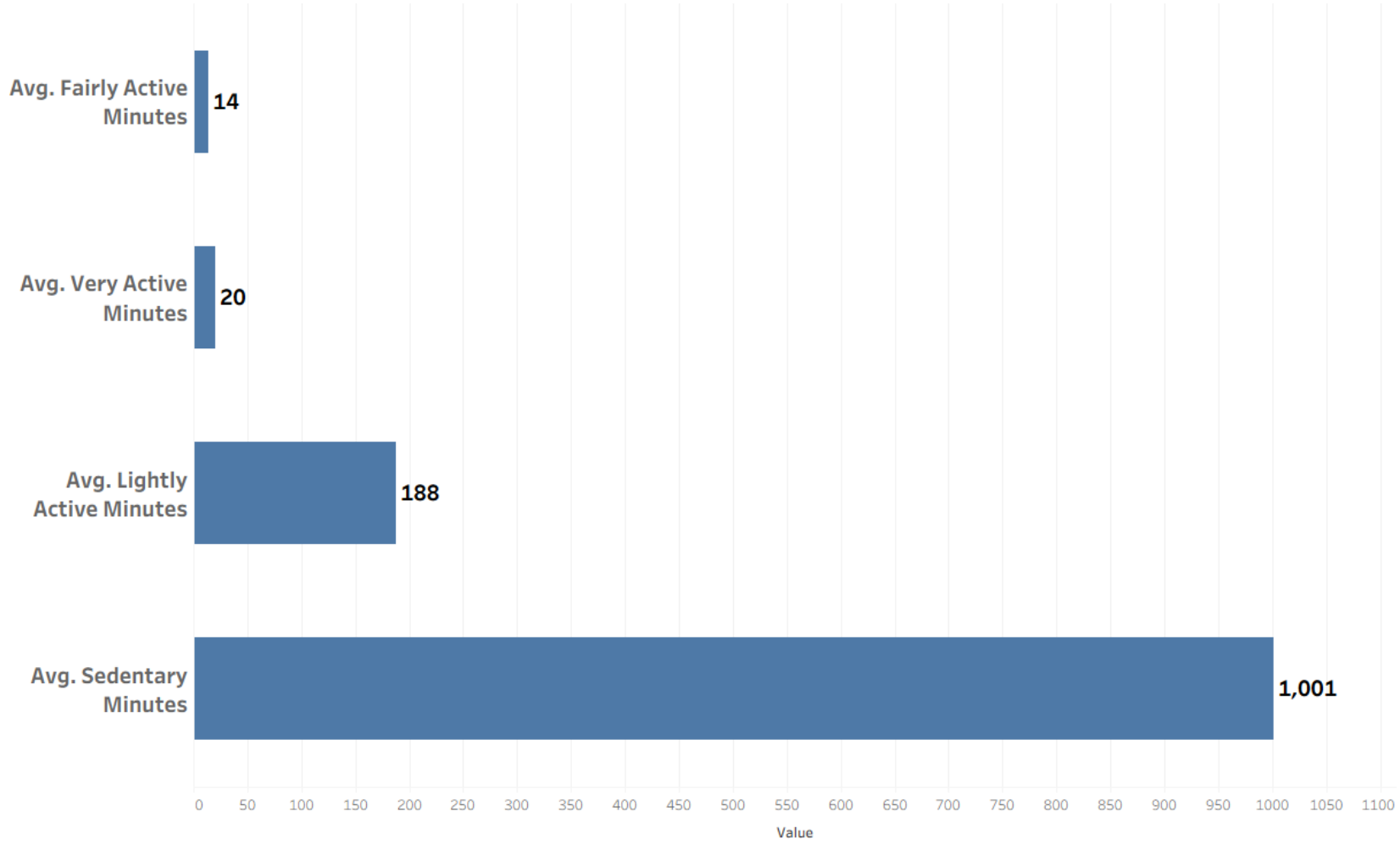
- Colored marks indicate dates with recorded data
- Gaps indicate dates without records

PARTICIPANT COVERAGE BY DATA TYPE

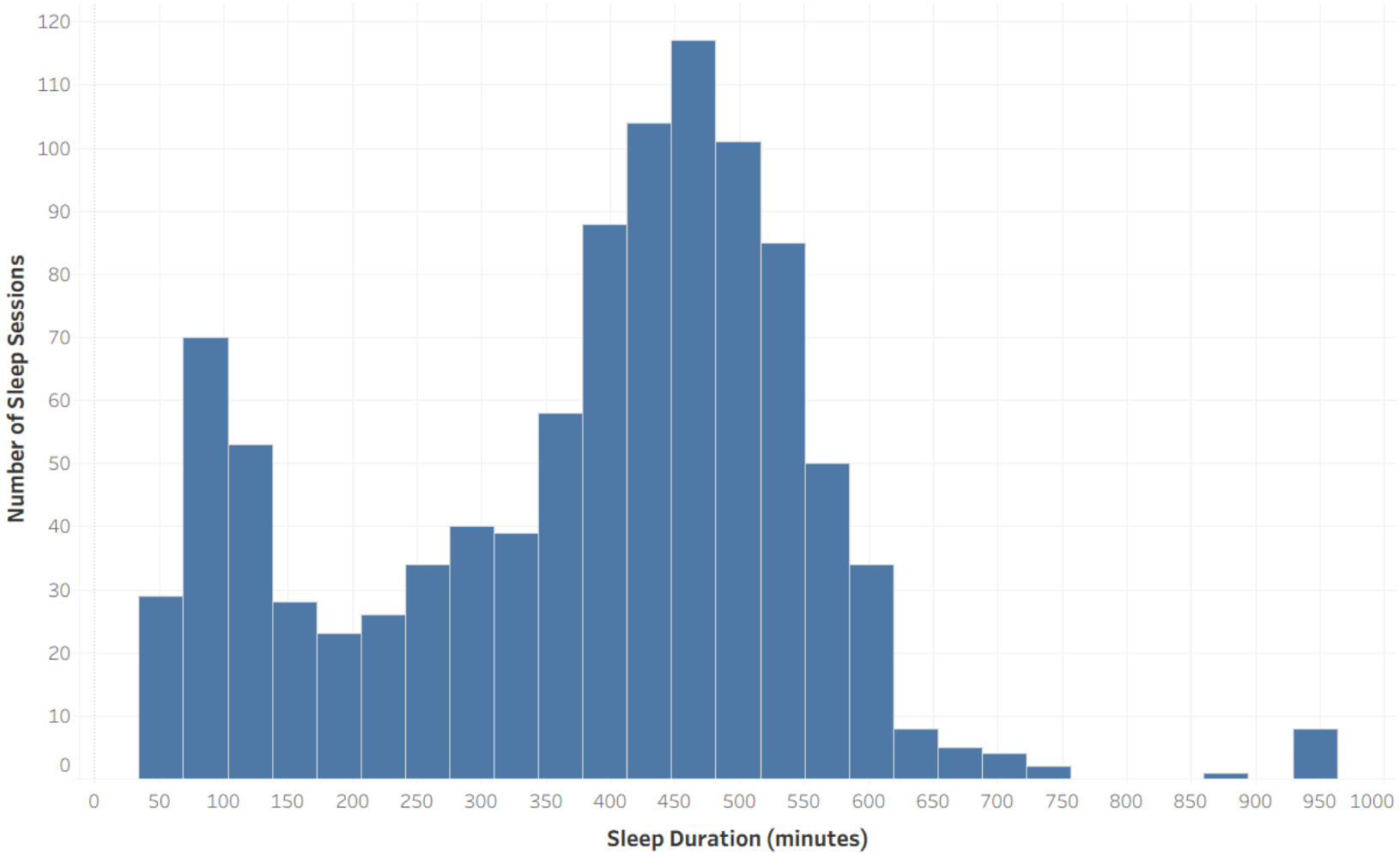
Manually logged activity is identified within daily activity records



AVERAGE DAILY MINUTES BY ACTIVITY INTENSITY



SLEEP DURATION DISTRIBUTION



MARKETING RECOMMENDATIONS FOR THE BELLABEAT APP



Given the substantial amount of recorded sedentary time, Bellabeat could explore marketing messages that emphasize incorporating movement into everyday routines.



Explore sleep awareness messaging

ACTIONABLE NEXT STEPS



Evaluate Bellabeat customer data to determine whether similar activity and sleep patterns are present



Assess customer interest in everyday movement and sleep awareness messaging



Test potential marketing messages before implementing a broader campaign

DATA SOURCES AND SUPPORTING ANALYSIS

- Data source: Fitbit Fitness Tracker Data – Kaggle
- Tableau Exploratory Visualizations
- Fitbit Sleep Data Analysis (Python/Jupyter Notebook)

GENERATIVE AI USE DISCLOSURE

ChatGPT was used as a learning and technical support tool throughout this project, including Python tutoring, code development and debugging assistance, troubleshooting, project organization, and clarification of analytical concepts.

Analytical questions, methodological decisions, interpretation of results, and final conclusions were independently evaluated and determined by the author. AI-generated suggestions and code were reviewed and tested by the author.